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YoungJo Choi (Jo Choi)

EDUCATION

New York University

- M.S., Computer Engineering

Fall 2024-Present

Expected graduation date: May 2026

Illinois Institute of Technology (Dual Degree program)

- B.S., Computer Engineering, minor in Artificial Intelligence

Fall 2022-Spring 2024

Inha University (Dual Degree program)

- B.S., Electronic Engineering

Spring 2016-Spring 2024

SKILLS

- **Programming&Scripting:** Python, C++, MySQL, C, Java, R, Verilog, SystemVerilog, VHDL, RISC-V Assembly, ARM Assembly, MATLAB
- **AI/ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Keras, MLflow
- **Data & Visualization:** Pandas, NumPy, PowerBI, Tableau, Excel, Selenium
- **Other:** React, HTML/CSS/JavaScript, Node.js, REST API, FastAPI, Flask, SageMaker AWS, Docker, Kubernetes, Linux, Git

WORK EXPERIENCE

Teaching Assistant at New York University

September 2025-December 2025

- Teaching lab sections for ECE-UY 3114 Fundamentals of Electronics I at New York University
- Ran experiments across amplifiers, diodes and transistor circuits

Data Analyst Intern at C&B Trading ((주)씨엔비트레이딩) in Seoul, Korea

June 2024-August 2024

- Improved sales forecast accuracy to 90% with a predictive model; built Excel reports and PowerBI dashboards that identified three key process bottlenecks for senior management.

Republic of Korea (ROK) Military

Aug 2017-April 2019

- Honorably discharged as a Sergeant and a Squad Leader after serving in ROK for 21 months

RESEARCH EXPERIENCE & PROJECTS

Efficient Vision-Language Model (VLM) Design with Sparse techniques

September 2025-Present

- 1) Developing a saliency-based Vision-Language Model (VLM) by integrating a Vision Transformer with an LLM through a projection layer, enabling efficient visual token selection and multimodal reasoning under the guidance of Prof. Sai Qian Zhang
- 2) Applied SparseVLM technique to Llava model and successfully reduced computation by 20%

GenAI-based Chip Design

Fall 2025

- 1) Developing LLM model architectures for automating chip design across RTL, verification to physical design, also for design optimization under the guidance of Prof. Ramesh Karri
- 2) Designing a Quantized Neural Net Accelerator with the Microwatt OpenPOWER CPU on SKY130, enabling efficient ML inference via memory-mapped I/O and custom instruction paths

OpenPOWER SoC Design with Microwatt CPU and Neural Network Accelerator

Fall 2025

- FPGA/ASIC-ready neural network accelerator (Verilog RTL) featuring INT8 systolic array processing for OpenPOWER systems

Superconducting Nanoribbon Device Fabrication

March 2025-August 2025

- Developed superconducting nanoribbon wires on HbN substrate at Nano Lab under the guidance of Prof. Davood Shahrjerdi
- Engineered the recipes of E-beam lithography, Deep Reactive-Ion Etching, and Buffered Oxide Etching processes in cleanroom environment

Mouse Motion Tracker using Deep Learning at Illinois Institute of Technology

Sep 2023-May 2024

- Developed an RNN-based multimodal system using visual and audio data to recognize six mouse behaviors with 90% accuracy under the guidance of Prof. Yan Yan

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AWARDS & SCHOLARSHIPS

Dean's List at IIT

Spring 2023

Honor Student at Inha University & Merit Scholarship

Fall 2019

LEADERSHIP EXPERIENCE

Active member at **Triangle fraternity (Armour Chapter)**

Fall 2022-Spring 2024

President & Founder of Korean Student Association at Illinois Institute of Technology Summer 2023-Spring 2024